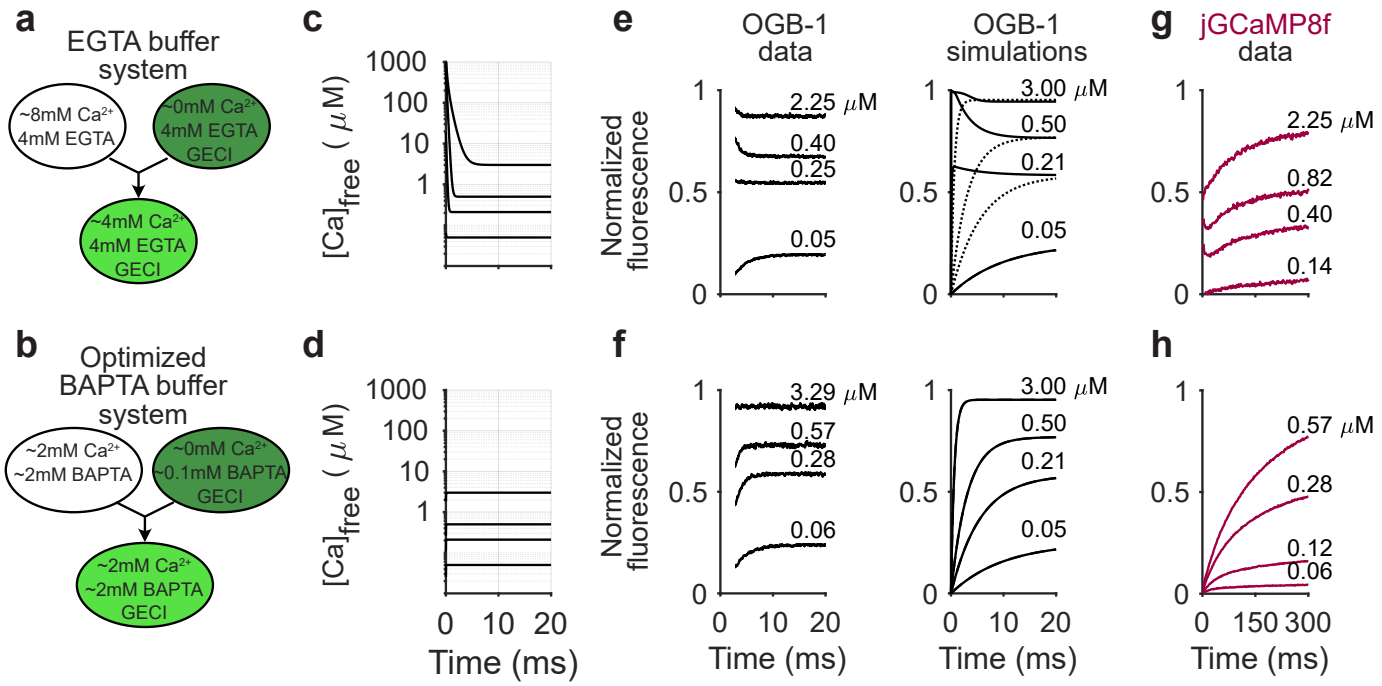
Supplementary Information

Precise calcium-to-spike inference using biophysical generative models

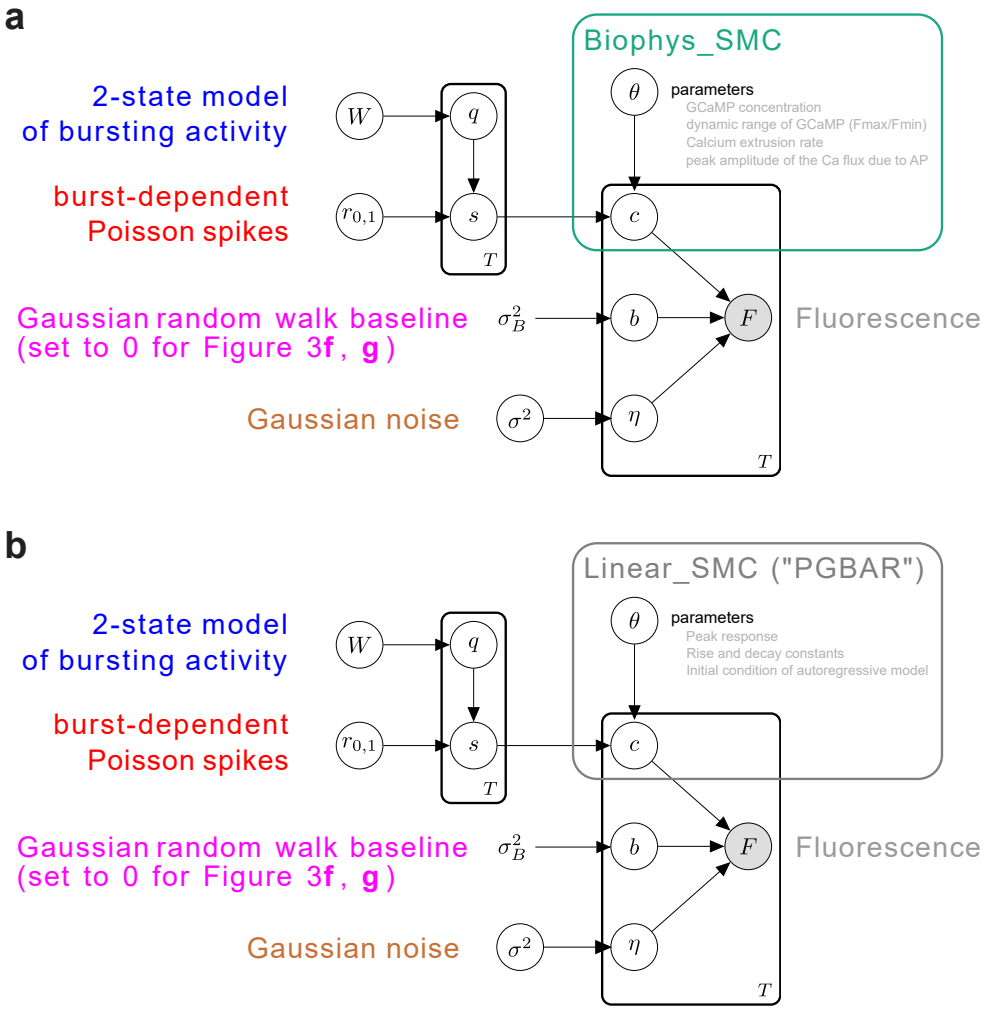
Supplementary Figures 1-4, Supplementary Tables 1-3, Supplementary Notes 1-2 and Supplementary References

Supplementary Figure 1



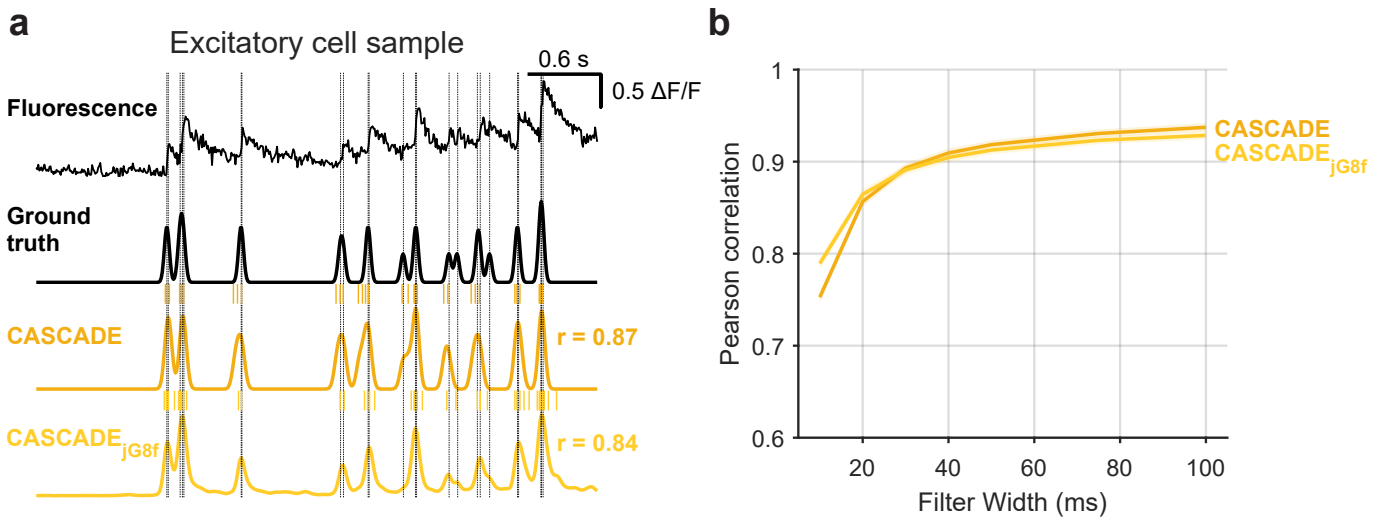
The impact of calcium clamp design on observed stopped-flow dynamics. (a-b) Schematic of stopped-flow buffer systems using **(a)** a slow buffer (EGTA) and containing unbuffered calcium in the calcium-containing bolus and **(b)** a system with a fast buffer (BAPTA) and fully buffered calcium in the calcium-containing bolus. **(c)** Simulations of the EGTA-based system presented in **(a)** show uncontrolled free calcium concentrations in the first few milliseconds of large calcium steps, but **(d)** BAPTA-based experiments do not. **(e)** Measurements of OGB-1 responses (left) reflect uncontrolled responses that are matched by simulations of the buffer system (right). **(f)** OGB-1 data and simulations in BAPTA-based buffers. **(g)** Artifactual fast on-response to jGCaMP8f in EGTA-based buffer system. **(h)** True jGCaMP8f response in BAPTA-based buffer system.

Supplementary Figure 2



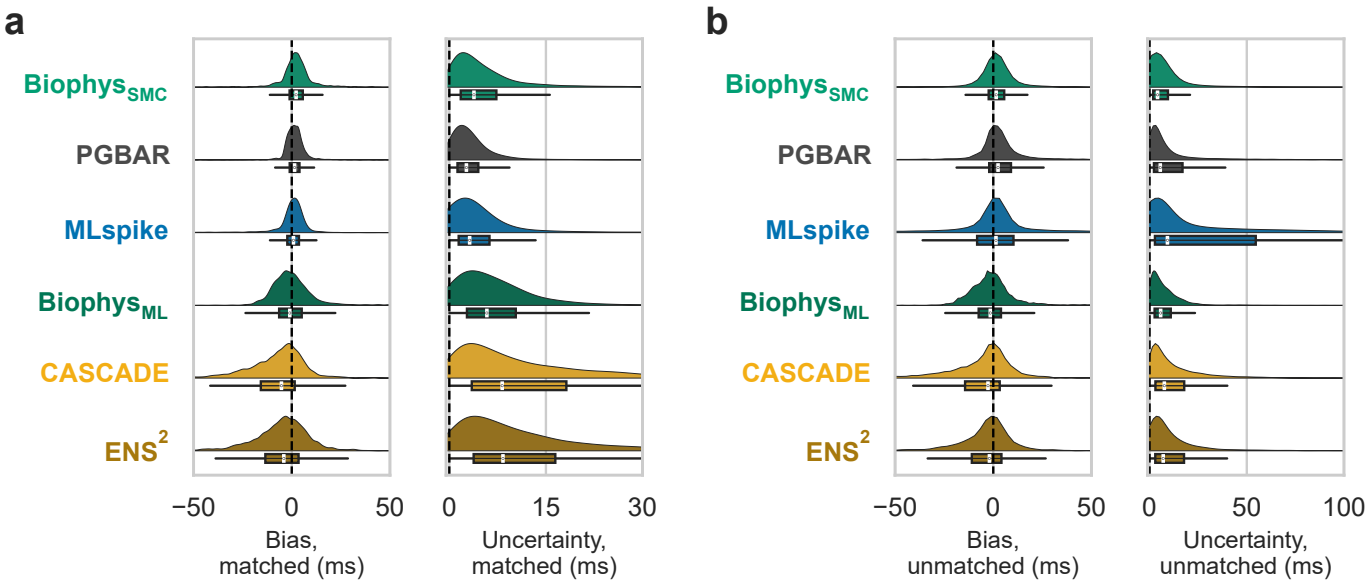
Generative cell-based model of fluorescence time series (a) Graphical representation of the generative SMC model described in the main text. White circles denote unknown variables, gray circles denote measurements and the bare variable σ^2_B is a fixed prior hyperparameter controlling the Brownian motion of the fluorescence baseline. This value was held fixed at 0 for the demonstration in Figure 3**f**, **g**. Rectangular outlines denote groups of variables. Note that GCaMP model parameters are included, but other than concentration are fixed for the SMC approach. **(b)** The key change over Linear_SMC/PGBAR (Diana et al., 2026) is the replacement of the autoregressor model with a biophysically-inspired parametric model (green rectangle in BiophysicalSMC, gray in Linear_SMC/PGBAR).

Supplementary Figure 3



Training on jGCaMP8f data does not improve CASCADE performance. (a) Imputed spike probabilities using the CASCADE Global Exc model or one trained on the jGCaMP8f dataset itself. Values of r indicate Pearson correlation between the time course of ground truth and that of the imputed probabilities for data filtered at 20 ms. Vertical lines indicate true spike times and tick marks indicate imputed spikes. **(b)** Mean Pearson correlations across excitatory recording epochs for different filtering time constants; shaded bands indicate ±SEM.

Supplementary Figure 4



Comparison of matched and unmatched temporal differences across methods. (a) Bias and uncertainty between imputed and ground truth spikes matched within a 100 ms window. Each plot is a kernel density estimate over a boxplot of the indicated metric for each imputed spike. Boxes show medians and 25th–75th percentiles; whiskers terminate at the most extreme observed values within 1.5 IQR of the box edges. Caps and separate outlier symbols are omitted; all finite source values contribute. **(b)** As **a** for all imputed spikes compared to the nearest ground truth spike. Note that the supervised methods show increased bias and uncertainty when removing double-counted imputed spikes by using matched statistics. Across 37 cells, **a** contains 4,941, 5,345, 3,302, 4,538, 4,807 and 3,757 matched per-spike observations for BiophysSMC, PGBAR, MLspike, BiophysML, CASCADE and ENS2, respectively; **b** contains 5,933, 9,937, 7,244, 5,536, 5,746 and 4,157 unmatched observations in the same order. These observations are individual spikes nested within recorded epochs and cells.

Supplementary Table 1

Parameter (units)	Cell Type	Value	References
GCaMP Parameters			
[GCaMP] (μM)	Both	50	Zariwala et al., 2012; Huber et al., 2012; Rose et al., 2014
R_f (unitless)	Both	5	This study
Calcium Handling			
ΔCa_{spike} (μM)	Both	70	Helmchen and Tank, 2015
ca_0 (μM)	Both	0.05	Maravall et al., 2000
γ (s^{-1})	Both	1800	Brenowitz and Regehr, 2007
Endogenous Buffer			
$Buffer_{Endo}$ (μM)	Both	12000	This study
$k_{on, Endo}$ (μM^{-1} s^{-1})	Both	12000	Xu et al., 1997
$k_{off, Endo}$ (s^{-1})	Both	10000	Xu et al., 1997
ATP Ca^{2+} Binding and Concentration			
[ATP] (μM)	Both	2000	Rebola et al., 2019
$k_{on, ATP}$ (μM^{-1} s^{-1})	Both	500	Naraghi & Neher, 1997
$k_{off, ATP}$ (s^{-1})	Both	100000	Naraghi & Neher, 1997
Parvalbumin Ca^{2+} Binding Properties			
[Parv] (μM)	Interneuron	1000	Eggermann and Jonas, 2012
$k_{on, Parv, Ca^{2+}}$ (μM^{-1} s^{-1})	Interneuron	400	Mueller et al., 2007
$k_{off, Parv, Ca^{2+}}$ (s^{-1})	Interneuron	4	Mueller et al., 2007
$k_{on, Parv, Mg^{2+}}$ (μM^{-1} s^{-1})	Interneuron	1	Mueller et al., 2007
$k_{off, Parv, Mg^{2+}}$ (s^{-1})	Interneuron	30	Mueller et al., 2007
mg_0 (μM)	Interneuron	700	Romani and Scarpa, 1992

Single-cell model parameter values.

Supplementary Table 1. Parameter definitions

[GCaMP] Total intracellular indicator concentration used in the model.

R_f Apparent fluorescence dynamic range (scaling factor mapping indicator state to measured ΔF/F).

ΔCa_{spike} Per-spike calcium increment delivered to the modeled compartment.

ca_0 Basal intracellular calcium concentration.

γ Effective first-order calcium clearance rate constant.

$Buffer_{Endo}$ Total endogenous calcium buffer concentration.

$k_{on,\ Endo}$ Calcium binding on-rate to the endogenous buffer.

$k_{off,\ Endo}$ Calcium unbinding off-rate from the endogenous buffer.

[ATP] Total ATP concentration.

$k_{on,\ ATP}$ Calcium binding on-rate to the ATP buffer.

$k_{off,\ ATP}$ Calcium unbinding off-rate from the ATP buffer.

[Parv] Total parvalbumin concentration.

$k_{on,\ Parv,\ Ca^{2+}}$ Calcium binding on-rate to the parvalbumin buffer.

$k_{off,\ Parv,\ Ca^{2+}}$ Calcium unbinding off-rate from the parvalbumin buffer.

$k_{on,\ Parv,\ Mg^{2+}}$ Magnesium binding on-rate to the parvalbumin buffer.

$k_{off,\ Parv,\ Mg^{2+}}$ Magnesium unbinding off-rate from the parvalbumin buffer.

mg_0 Basal intracellular magnesium concentration.

Supplementary Table 2

Parameter	Description	Units	GCaMP6f	jGCaMP7f	jGCaMP8f Janelia	jGCaMP8f Slice	jGCaMP8m
K_{aN}	Affinity [N-term]	μM	0.873	1.69	0.414	0.514	0.323
k_{offN}	Off Rate [N-term]	s^{-1}	7.4	43.1	25.2	56.5	111
H_N	Hill coeff [N-term]	unitless	2.33	2.04	1.68	1.75	1.61
K_{aC}	Affinity [C-term]	μM	0.0404	0.119	0.00867	0.0239	0.139
k_{offC}	Off Rate [C-term]	s^{-1}	7.13	18.1	0.737	0.461	0.916
H_C	Hill coeff [C-term]	unitless	2.61	2.8	0.589	0.956	1.77
k_{onPN}	On Rate N [N-lobe bound]	s^{-1}	595	889	220	38.5	142
k_{offPN}	Off Rate N [N-lobe bound]	s^{-1}	292	134	210	345	96.4
k_{onPC}	On Rate C [C-lobe bound]	s^{-1}	2.79	0.261	10.6	0.136	3.37×10^{-6}
k_{offPC}	Off Rate C [C-lobe bound]	s^{-1}	796	4.51	172	36.8	83.9
k_{onPN2}	On Rate N [both bound]	s^{-1}	10	30.8	299	59.6	997
k_{offPN2}	Off Rate N [both bound]	s^{-1}	3.07	6.87	114	386	133
k_{onPC2}	On Rate C [both bound]	s^{-1}	155	134	2.01×10^{-4}	0.625	11.3
k_{offPC2}	Off Rate C [both bound]	s^{-1}	11.4	22.6	1.25	29.4	0.99

GCaMP model parameters sets.

Supplementary Table 3

Parameter (units)	Fixed/Free	Mean	Approximate range	References
GCaMP Parameters				
[GCaMP] (μM)	Free	60	1-150	Zariwala et al., 2012; Huber et al., 2012; Rose et al., 2014
R_f (unitless)	Free	5 (jGCaMP8f) 10 (jGCaMP8m)	3-7 (jGCaMP8f) 8-12 (jGCaMP8m)	This study
Calcium Handling				
ΔCa_{spike} (μM)	Free	15	1-100	Helmchen and Tank, 2015
ca_0 (μM)	Fixed	0.05		Maravall et al., 2000
γ (s^{-1})	Free	750	200-2500	Brenowitz and Regehr, 2007
γ_{in} (s^{-1})	Free	5	0.5-50	This study
γ_{out} (s^{-1})	Free	5	0.5-50	This study
Linearized Endogenous Buffer				
κ_B (unitless)	Fixed	40		Rebola et al., 2019

Cell-parameter prior means and approximate ranges.

Parameter definitions

[GCaMP] Total intracellular indicator concentration used in the model.

R_f Apparent fluorescence dynamic range (scaling factor mapping indicator state to measured ΔF/F).

ΔCa_{spike} Per-spike calcium increment delivered to the modeled compartment.

ca_0 Basal intracellular calcium concentration.

γ Effective first-order calcium clearance rate constant.

γ_{in} Rate constant governing calcium influx/exchange into the modeled intracellular compartment.

γ_{out} Rate constant governing calcium efflux/exchange out of the modeled intracellular compartment.

κ_B Linearized endogenous buffering factor applied to intracellular calcium dynamics.

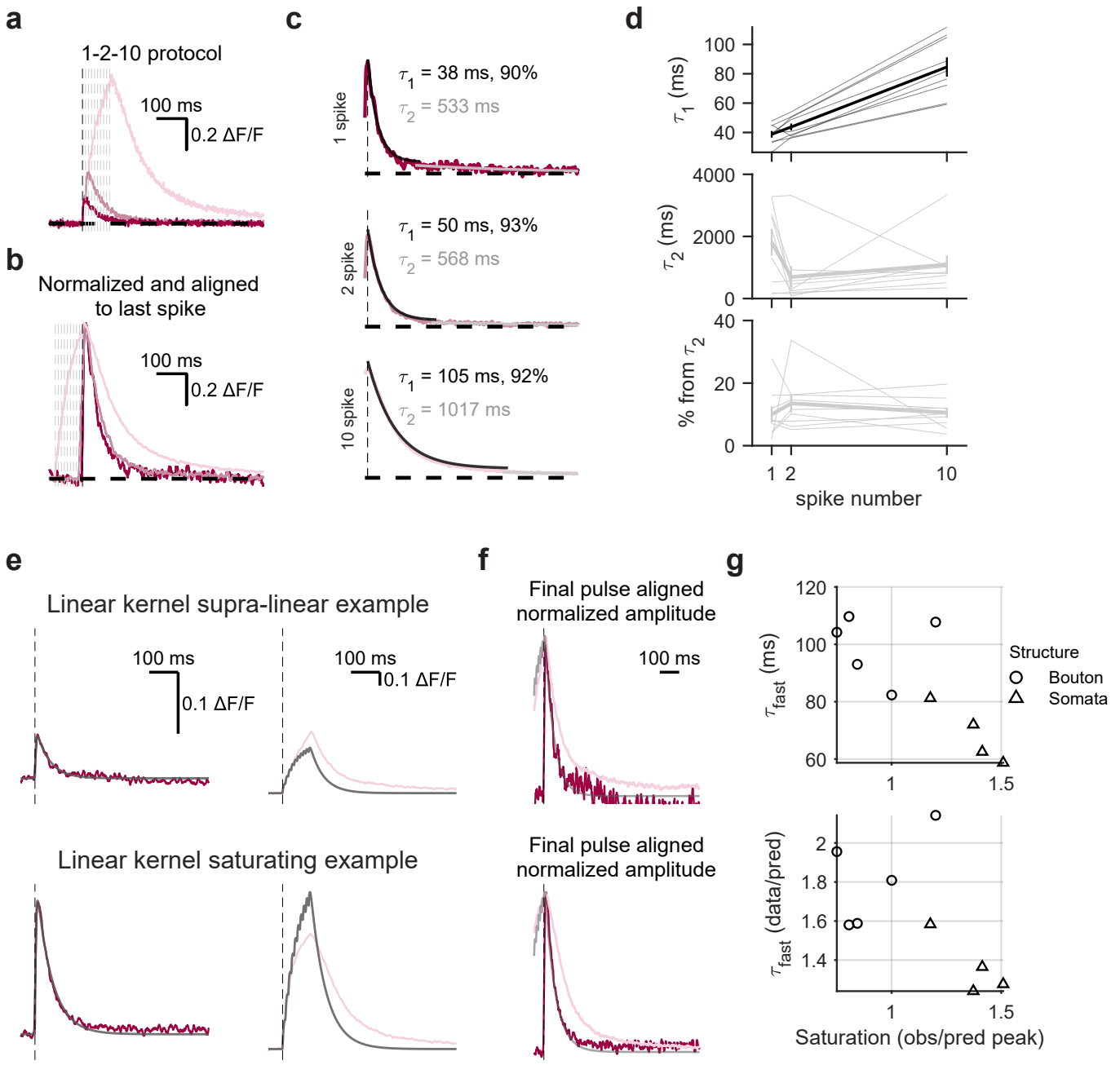
Supplementary Note 1. Activity-dependent decay slowing persists outside the saturating response regime

A reduction in fluorescence gain during repetitive stimulation could arise from probe-response saturation. Such amplitude compression, however, does not by itself establish that saturation accounts for the prolonged post-stimulus decay. We therefore compared the shapes of responses to 1, 2 and 10 stimuli after normalizing their amplitudes and aligning them to the final stimulus (panels **a**,**b**). Dual-exponential fits resolved fast and slow decay components, with the fast component becoming slower as stimulus number increased (panels **c**,**d**).

To distinguish this kinetic change from response saturation, we used the measured single-stimulus response to predict the response to a train by linear convolution at the known stimulus times. The examples span a non-saturating, supralinear response in a granule-cell soma and a saturating response in a bouton, assessed by comparing the observed train peak with the linear prediction (panel **e**). After normalization and alignment to the final stimulus, the ten-stimulus response decayed more slowly in both regimes (panel **f**). Thus, elongation was also present when the train response was not compressed relative to the linear prediction. The comparison across structures further shows decay slowing beyond that predicted by summation of the unitary response (panel **g**).

These controlled comparisons distinguish use-dependent changes in response kinetics from amplitude saturation: saturation may affect the amplitude of bouton responses, but cannot alone account for the observed elongation. The history-dependent decay therefore provides an additional constraint on a generative model of the spike-to-fluorescence transformation.

Figure accompanying Supplementary Note 1



jGCaMP8f response decay is captured by two kinetic components, while its slowing cannot be explained by response saturation. (a) Representative single-bouton responses to 1, 2, and 10 stimuli overlaid and aligned by first stimulus time. **(b)** Responses normalized and aligned to last spike. **(c)** Example fits to dual sums of exponentials showing fast and slow components of decay. **(d)** Dependence of fast (top) and slow (middle) time constants on the number of stimuli, and relative contribution of the slow component (τ_2) to the overall time course (bottom). Individual bouton values shown as low linewidth traces with group means shown as thicker line and standard error of the mean as error bars, n = 5 boutons and 4 granule cell somata. **(e)** Sample responses from granule cell soma (top) and bouton (bottom) with a linear fit to their respective single pulse response (left) and the convolution of the unitary response overlaid to the fluorescence response for each structure showing that GCaMP in the soma was operating in a non-saturating response regime while the bouton was beyond the linear regime. **(f)** Normalized alignment to the last pulse response in sub- (top) and supra-saturating regimes that demonstrate elongation of fluorescence response in the 10 pulse response in each case. **(g)** Fluorescence response elongation increases with cumulative calcium load as indexed by degree of supra- versus linear response (top) while the elongation is greater than predicted by the linear response (bottom).

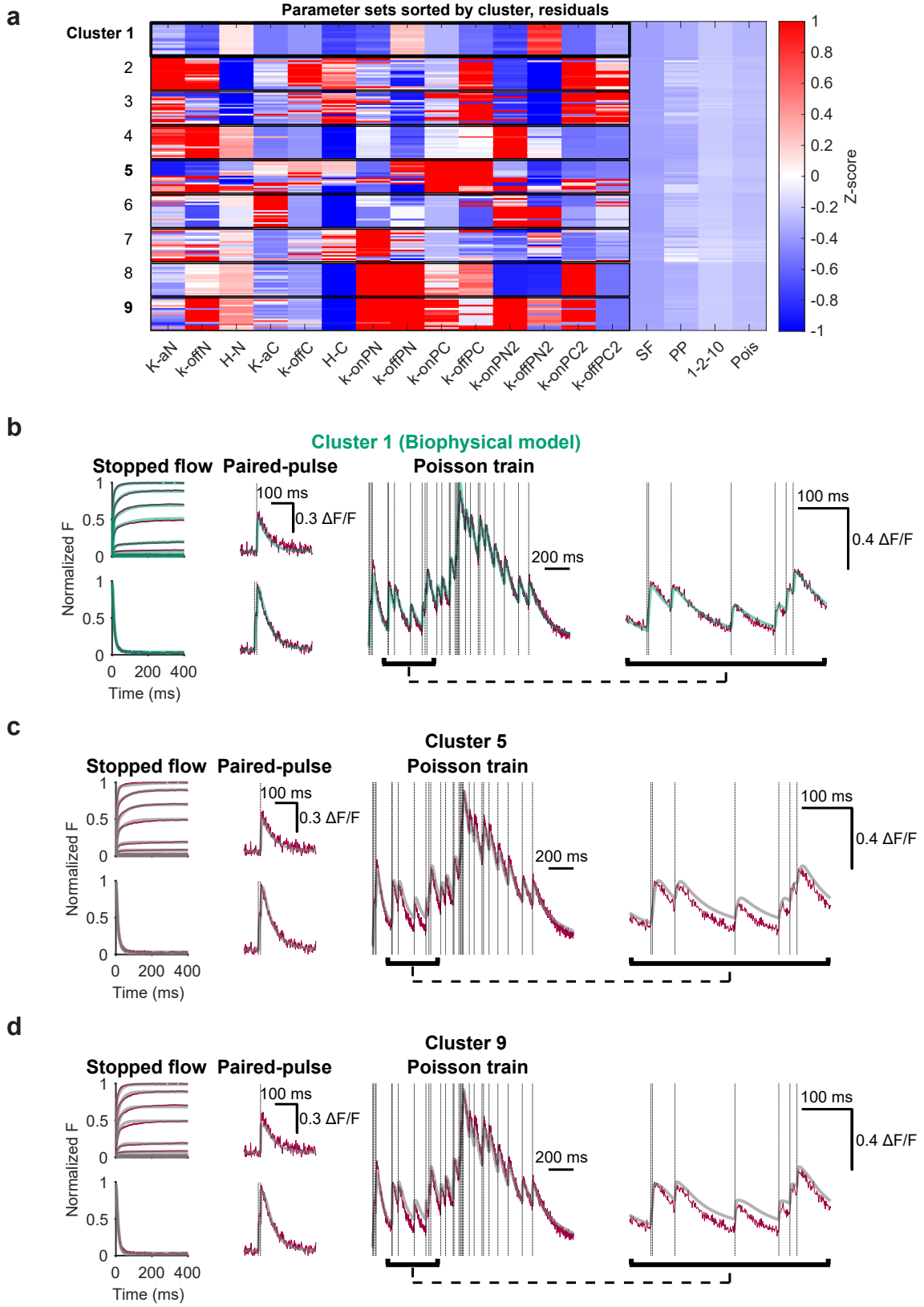
Supplementary Note 2. Activity-rich recordings distinguish candidate families of GCaMP model parameters

Fitting stopped-flow responses alone can leave multiple candidate parameter sets that describe the measured fluorescence kinetics. Our model-selection procedure therefore combined in vitro stopped-flow measurements with neuronal recordings containing defined stimulation patterns. Optimization from randomized starting parameters generated approximately 1,000 candidate GCaMP parameter sets. For each candidate, the GCaMP parameters were held fixed while the cell parameters were optimized against the neuronal responses, allowing the contribution of cell-specific calcium handling to be taken into account (Methods).

Grouping the fitted parameter sets by k-means into nine clusters made it possible to examine families of solutions associated with different regions of parameter space (panel **a**). The representative models from clusters 1, 5 and 9 illustrate why simple response regimes do not fully distinguish these families. Each gives a fit to the stopped-flow measurements and the paired-pulse examples, but their predictions diverge during the Poisson stimulation train (panels **b**–**d**). In particular, the examples from clusters 5 and 9 fail to capture the differential slowing of fluorescence across the train, visible on the expanded timescale. The representative model from cluster 1, selected for subsequent modeling, captures these activity-dependent features more closely.

The final model was selected using the combined stopped-flow and imaging constraints by minimizing the sum of the z-scored L2 norms of their fit residuals (Methods). The comparison demonstrates why recordings with varied activity patterns add information beyond a fit to stopped-flow responses or a brief paired-pulse protocol: they engage history-dependent sensor behavior and distinguish parameter families that appear similar under simpler conditions. Clustering here aids interpretation of candidate solutions; it does not by itself establish that every individual kinetic parameter is uniquely identifiable.

Figure accompanying Supplementary Note 2



Selection of jGCaMP8f biophysical parameters using in vitro and ex vivo datasets. (a) Distributions of parameters on fits from randomized seeds with GCaMP model parameters at figure left and residuals from best fits to stopped-flow and ex vivo experiments (paired pulse, 1-2-10 pulses, and Poisson train) based on optimizing cell parameters. GCaMP parameter sets were sorted by k-means clustering into nine clusters and then by lowest summed residuals within each cluster. The heatmap displays at most 20 candidate parameter sets per cluster. Each row represents a fitted parameter set, rather than an independent experimental replicate. This procedure allowed qualitative assessment of parameters that lay in different local minima of the parameter space. **(b)** Fitted data examples for cluster 1, which was used for all further modeling. **(c)** Fitted data for cluster 5. **(d)** Fitted data for cluster 9. In paired-pulse and Poisson train data, vertical lines indicate the times of action potentials. Expanded timescale of Poisson responses at right. Neither cluster 5 or 9 captured the differential slowing across the Poisson responses.

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